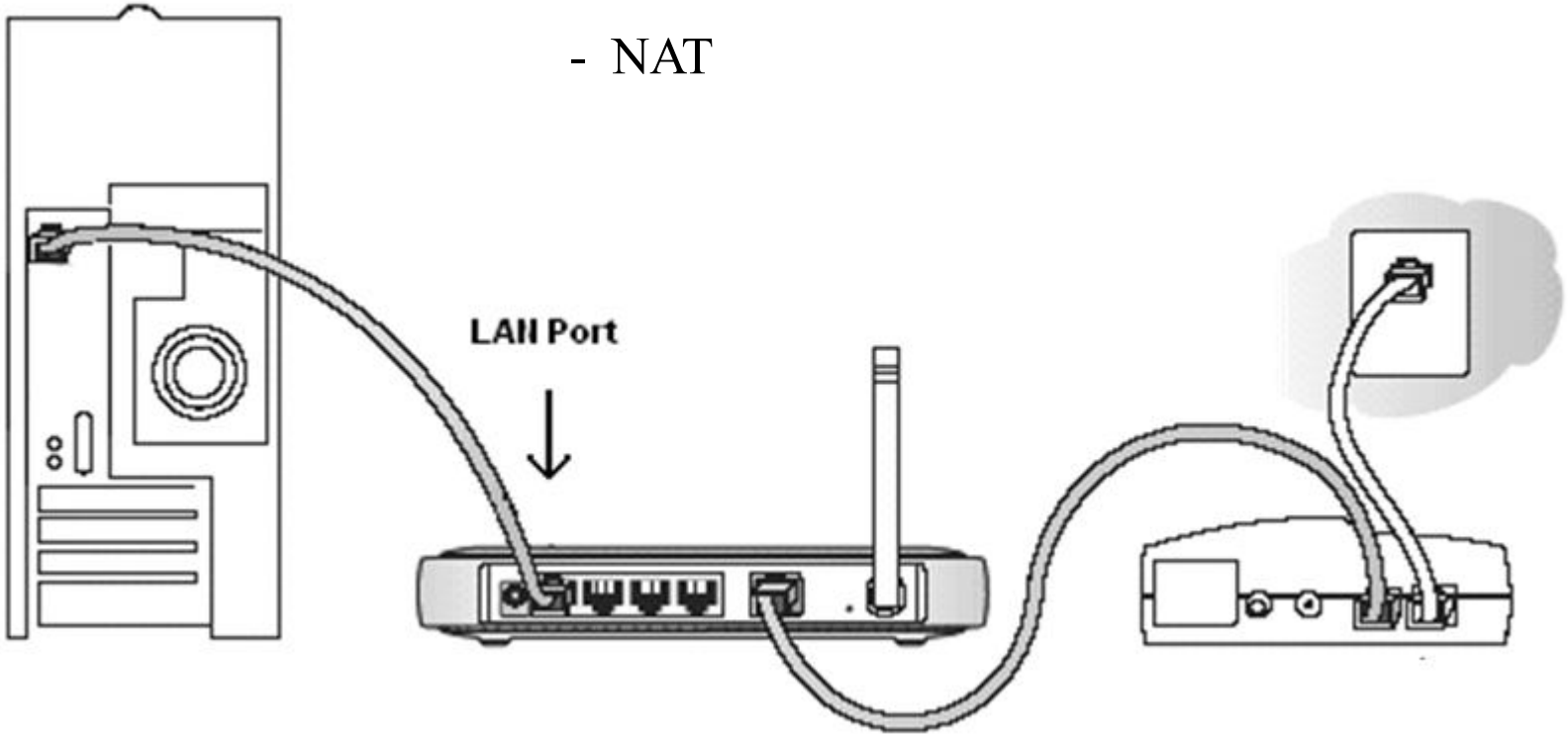
- 공유기 기능
  - 패킷 전달
  - DHCP
  - NAT



❶ 컴퓨터 ❷ 유/무선 공유기 ❸ 모뎀

사설/공인 IP Address

- **사설 IP 주소**
  - 인터넷과 연동되지 않은 사적인 독립 네트워크(Private IP Network)에서 사용되는 사적인 주소
  - 인터넷 상에서 사용할 수 없음을 의미

- Class 별 사설 IP 주소 대역(IETF RFC 1918)

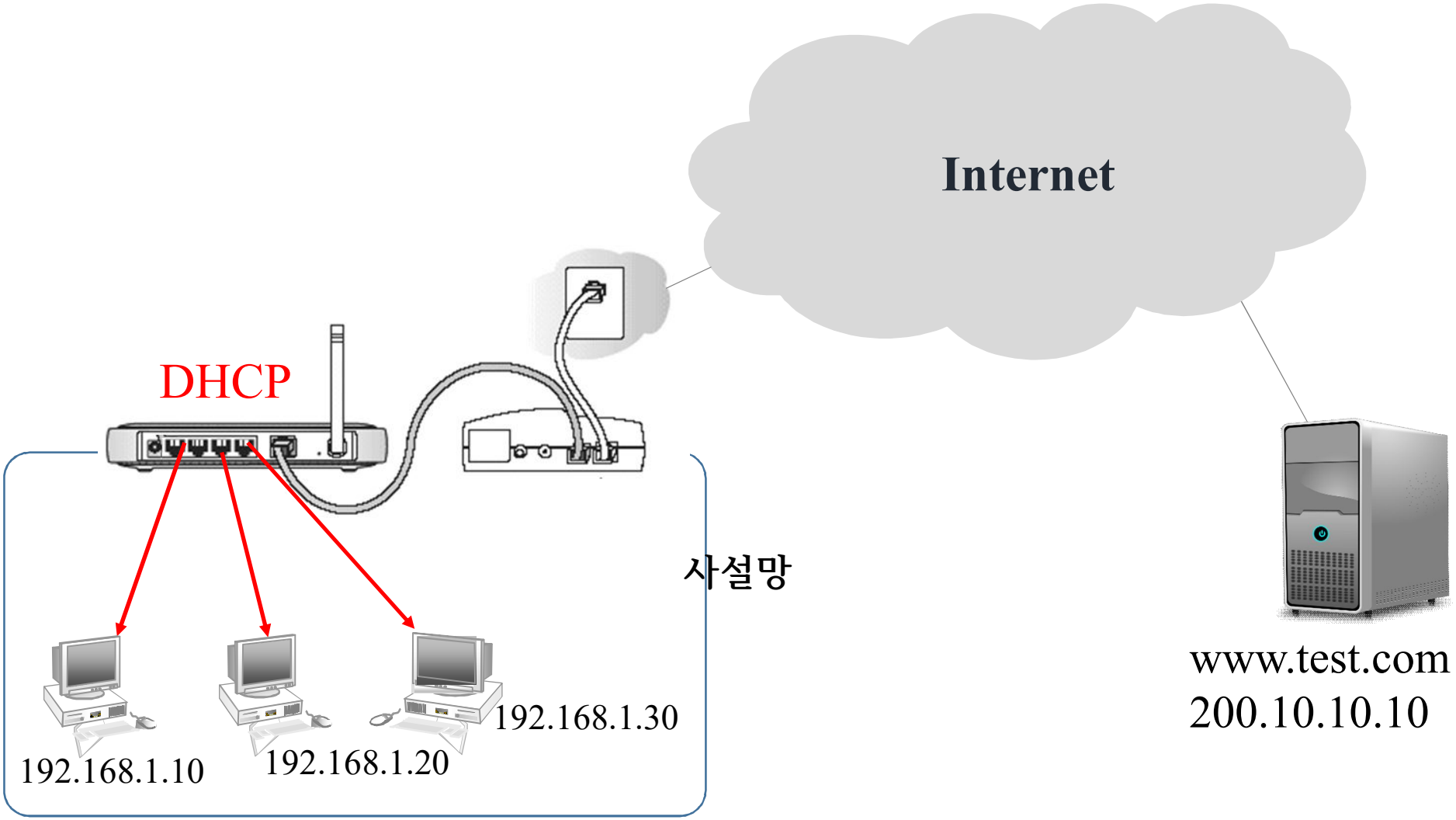
  ① A Class : 10.x.x.x

  ② B Class : 172.16.x.x ~ 172.31.x.x

  ③ C Class : 192.168.x.x

- **공인 IP주소**

  - 사설 IP주소를 제외한 주소


Internet
DHCP
사설망
192.168.1.10
192.168.1.20
192.168.1.30
www.test.com
200.10.10.10

NAT 기능

- 네트워크 주소 변환 (Network Address Translation)
- 송신지 또는 수신지 IP address를 다른 주소로 변환
- Source NAT(SNAT)와 Destination NAT(DNAT)

송신지 IP	수신지 IP
192.168.1.10	200.10.10.10

❶ **SNAT**

172.16.1.10	200.10.10.10

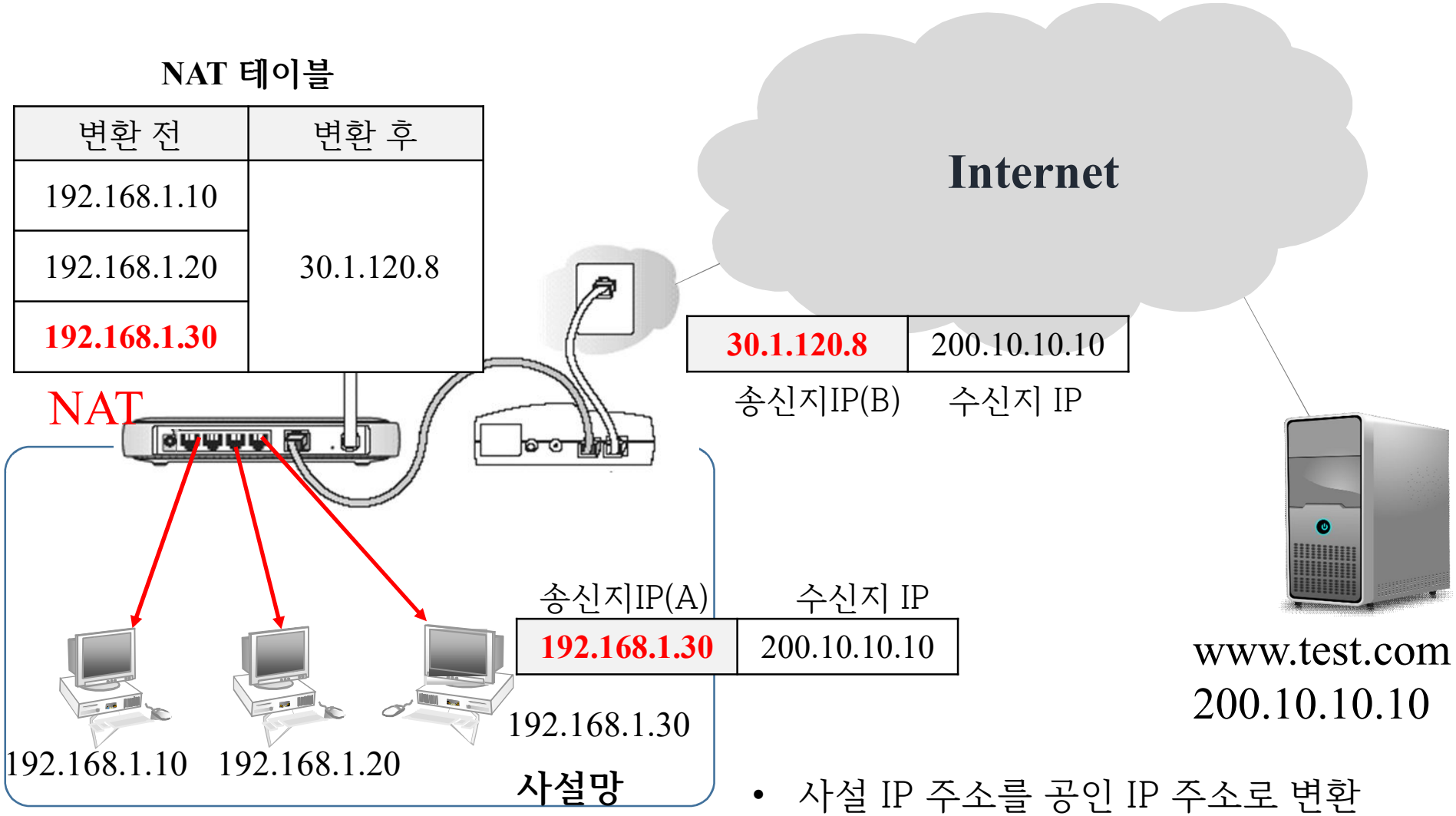
송신지 IP	수신지 IP
192.168.1.10	**200.10.10.10**

❷ **DNAT**

192.168.1.10	**10.10.10.10**

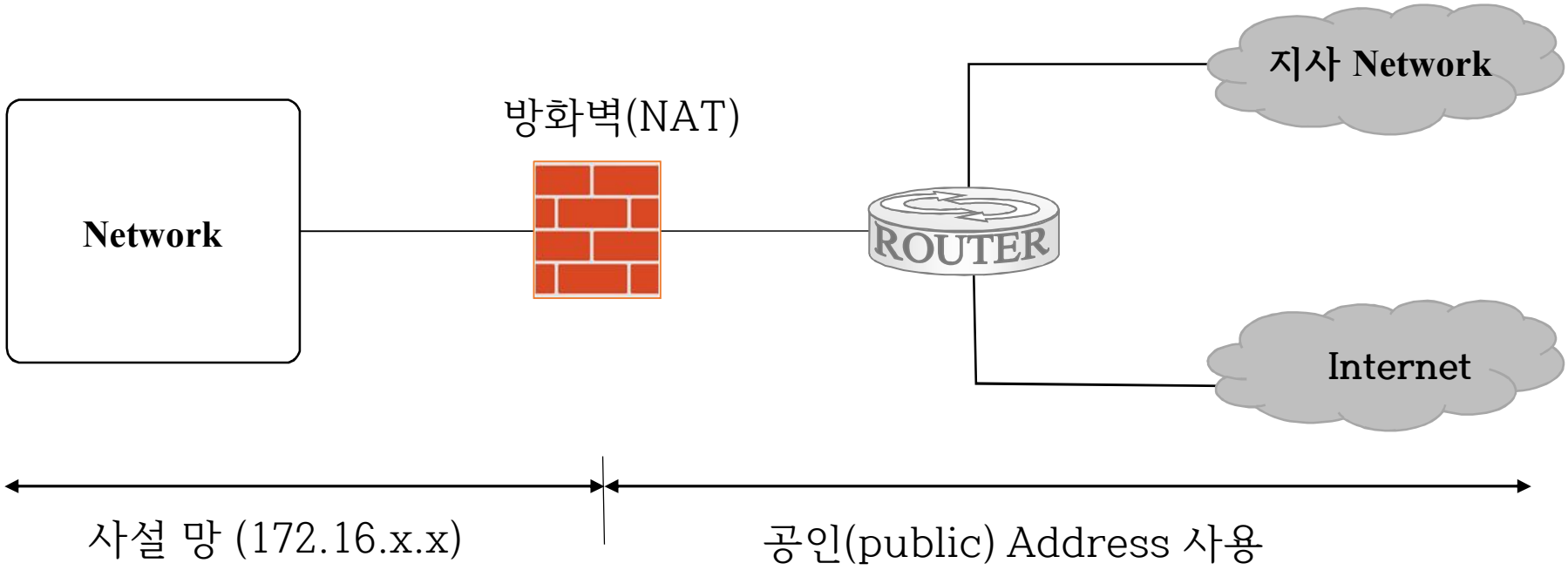
NAT 테이블

변환 전	변환 후
192.168.1.10	30.1.120.8
192.168.1.20	
192.168.1.30	



- 사설 IP 주소를 공인 IP 주소로 변환

기업망에서의 NAT

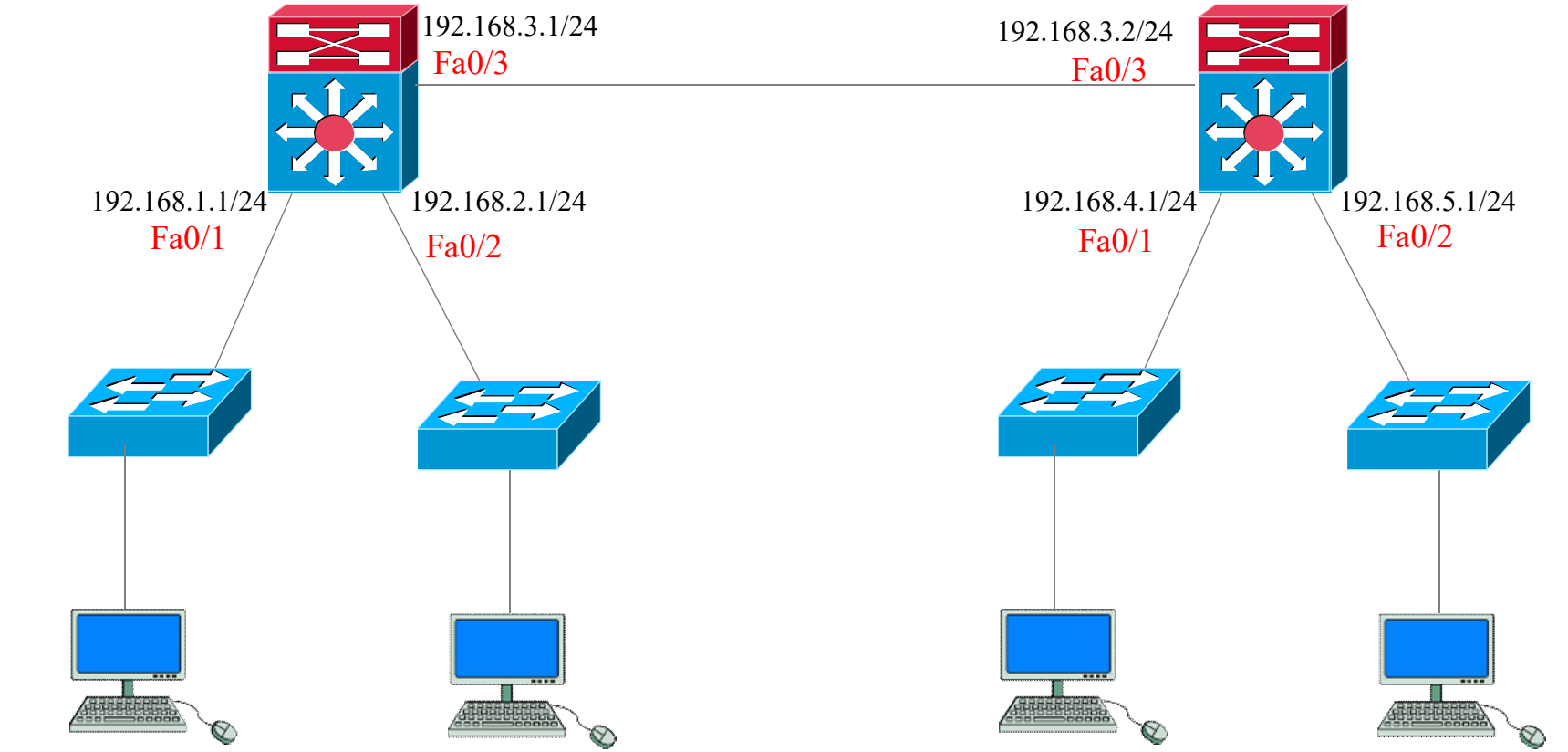

지사 Network
방화벽(NAT)
Network
ROUTER
Internet
사설 망 (172.16.x.x)
공인(public) Address 사용

Routing Table

192.168.1.0 fa0/1
192.168.2.0 fa0/2
192.168.3.0 fa0/3
192.168.4.0 fa0/3
192.168.5.0 fa0/3

Routing Table

192.168.1.0 fa0/3
192.168.2.0 fa0/3
192.168.3.0 fa0/3
192.168.4.0 fa0/1
192.168.5.0 fa0/2

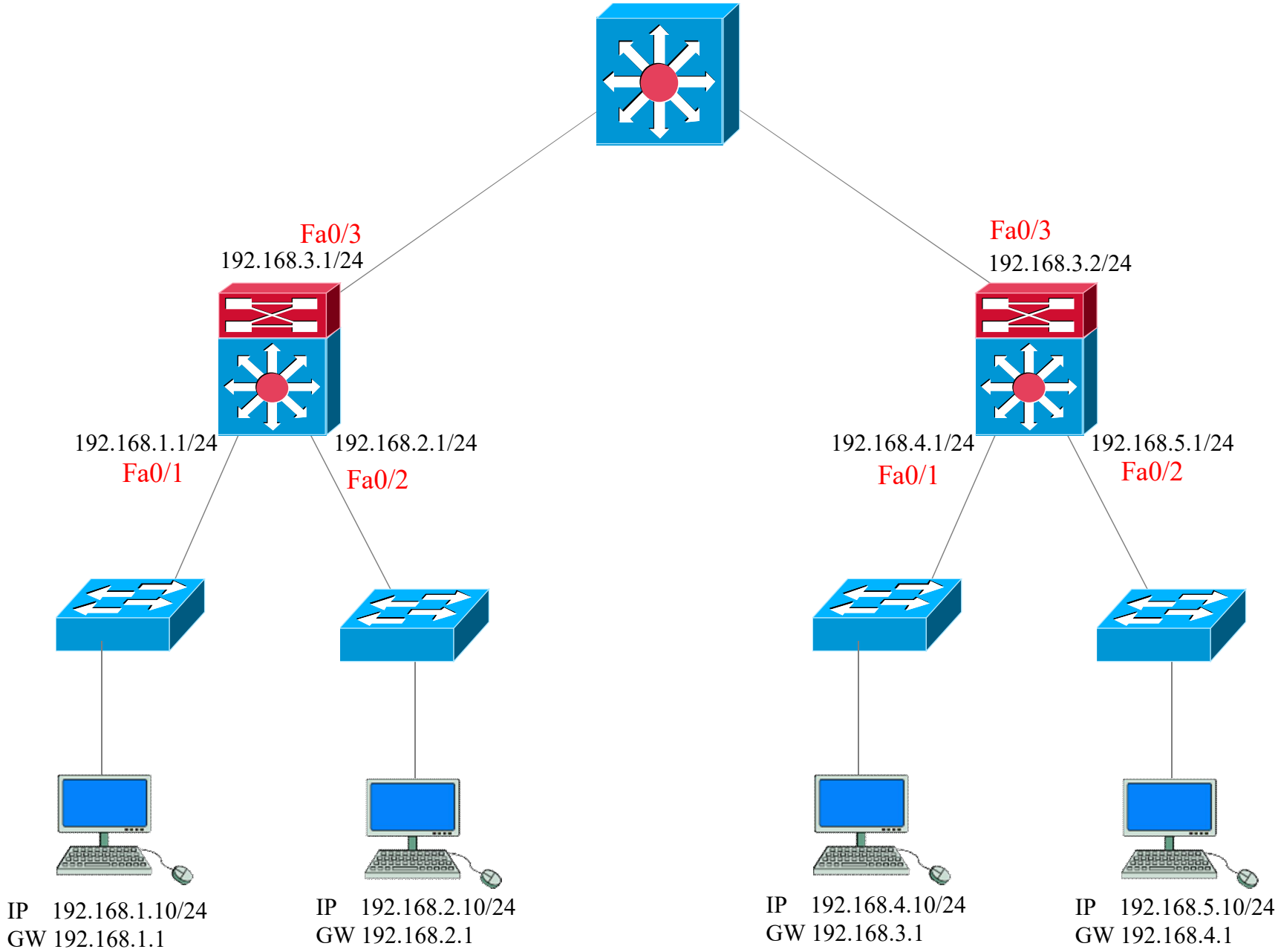


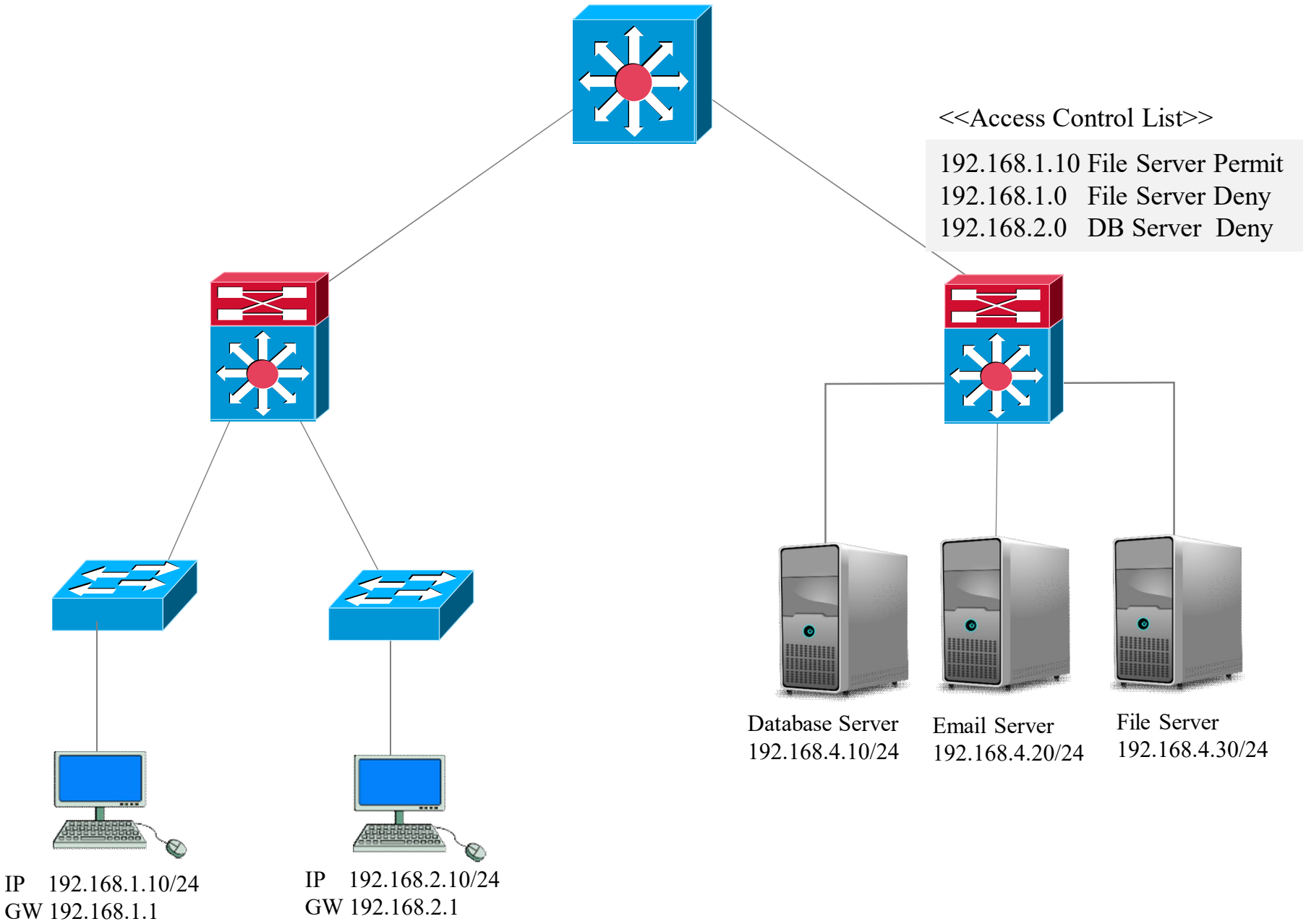
IP 192.168.1.10/24
GW 192.168.1.1

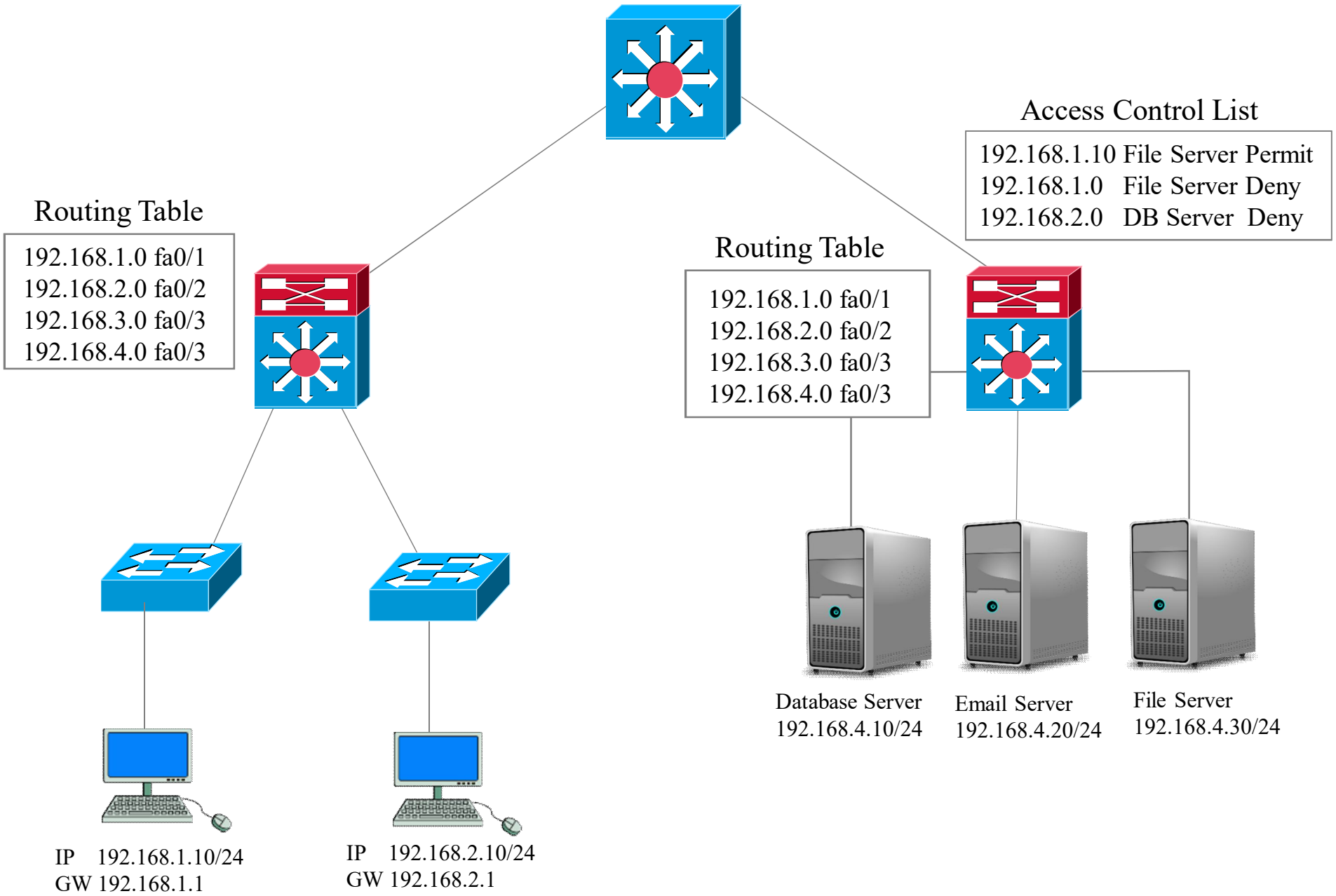
IP 192.168.2.10/24
GW 192.168.2.1

IP 192.168.4.10/24
GW 192.168.3.1

IP 192.168.5.10/24
GW 192.168.4.1

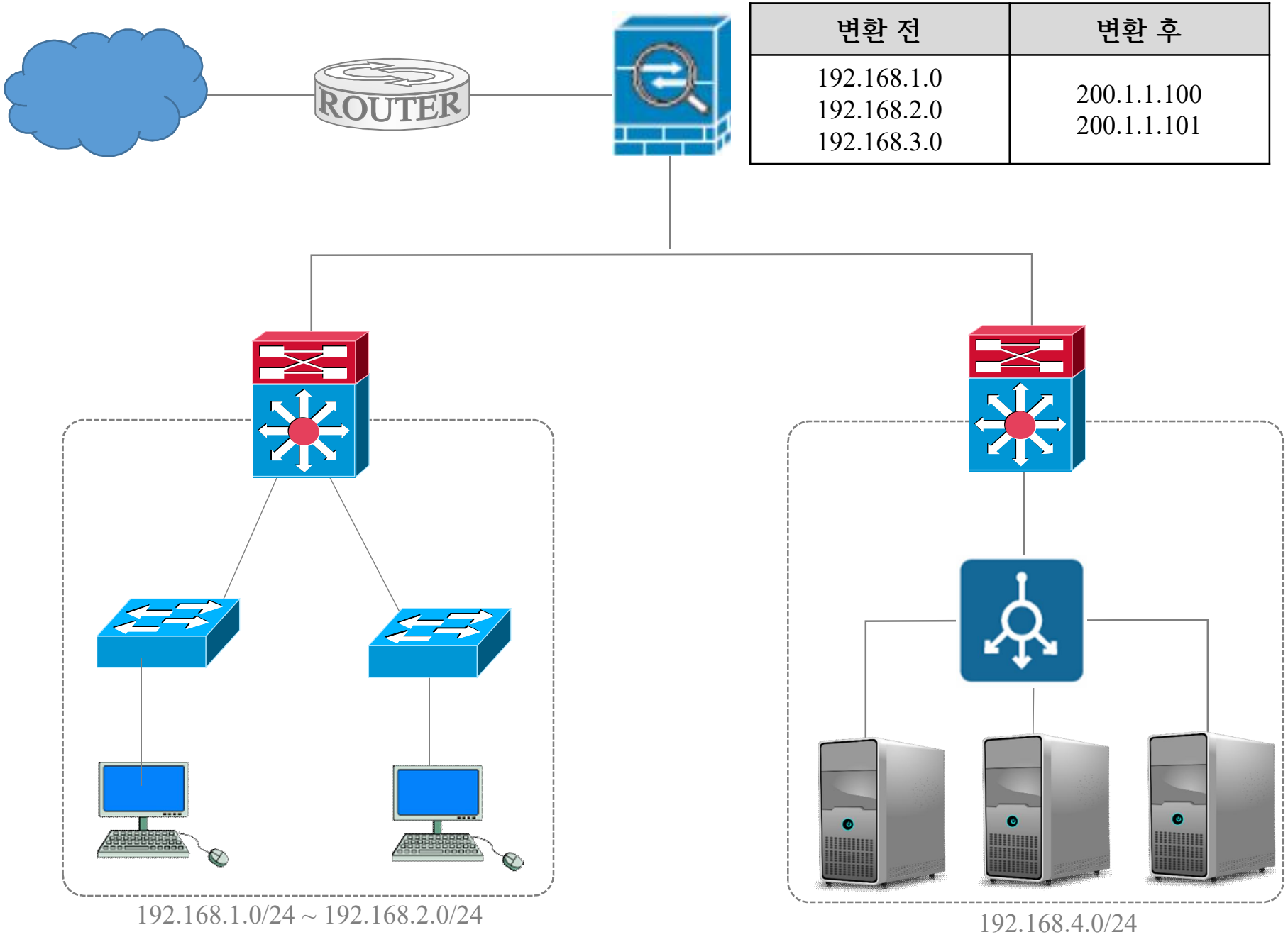

Fa0/3
192.168.3.1/24
Fa0/3
192.168.3.2/24
192.168.1.1/24
Fa0/1
192.168.2.1/24
Fa0/2
192.168.4.1/24
Fa0/1
192.168.5.1/24
Fa0/2
IP 192.168.1.10/24
GW 192.168.1.1
IP 192.168.2.10/24
GW 192.168.2.1
IP 192.168.4.10/24
GW 192.168.3.1
IP 192.168.5.10/24
GW 192.168.4.1


<<Access Control List>>
192.168.1.10 File Server Permit
192.168.1.0 File Server Deny
192.168.2.0 DB Server Deny
Database Server
192.168.4.10/24
Email Server
192.168.4.20/24
File Server
192.168.4.30/24
IP 192.168.1.10/24
GW 192.168.1.1
IP 192.168.2.10/24
GW 192.168.2.1


Access Control List
192.168.1.10 File Server Permit
192.168.1.0 File Server Deny
192.168.2.0 DB Server Deny
Routing Table
192.168.1.0 fa0/1
192.168.2.0 fa0/2
192.168.3.0 fa0/3
192.168.4.0 fa0/3
Routing Table
192.168.1.0 fa0/1
192.168.2.0 fa0/2
192.168.3.0 fa0/3
192.168.4.0 fa0/3
Database Server
192.168.4.10/24
Email Server
192.168.4.20/24
File Server
192.168.4.30/24
IP 192.168.1.10/24
GW 192.168.1.1
IP 192.168.2.10/24
GW 192.168.2.1

NAT Table

변환 전	변환 후
192.168.1.0 192.168.2.0 192.168.3.0	200.1.1.100 200.1.1.101



Router/Switch

- 브로드캐스트 주소와 멀티캐스트 주소를 등록하지 않음

Switch	브로드캐스트/멀티캐스트 트래픽 ➔ Flooding
Router	브로드캐스트/멀티캐스트 트래픽 ➔ Drop